

Linear algebra – 12

Mixed problems: commutators, traces and the numerical range

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Theoretical minimum.

Definition. The commutator is $[X, Y] = XY - YX$; the numerical range of $A \in M_n(\mathbb{C})$ is $W(A) = \{\langle Ax, x \rangle : \|x\| = 1\}$. The trace form $\langle X, Y \rangle = \text{tr}(XY^\top)$ makes $M_n(\mathbb{R})$ a Euclidean space, and most statements about “all matrices X ” are statements about that inner product.

Theorem 1 (Schur). Every complex matrix is unitarily similar to an upper triangular one, with the eigenvalues on the diagonal in any prescribed order. This is the standard first move when a problem mixes a spectral hypothesis with an inequality.

Theorem 2 (Shoda, Albert–Muckenhoupt). Over any field, A is a commutator $[X, Y]$ if and only if $\text{tr } A = 0$. In particular $[X, Y] = E$ is impossible, whereas a commutator of rank one does exist.

Fact 1 (Toeplitz–Hausdorff). $W(A)$ is a convex compact set containing the spectrum; for a normal A it is the convex hull of the eigenvalues. Convexity of $W(A)$ is the matrix counterpart of the convexity statements in Convex – 1...3.

Fact 2 (four moves for a problem with no visible topic). Apply the identity to a well-chosen vector — an eigenvector, or one from the kernel; take traces, which kills commutators; pass to 2×2 or to the Jordan form and check blockwise; or read the hypothesis as a statement about the linear map $X \mapsto AX - XB$ on M_n . If the matrices are real and the answer mentions signs, remember that non-real eigenvalues come in conjugate pairs and contribute positive factors to the determinant.

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1. (Romania national olympiad 2024) Let $A \in M_n(\mathbb{R})$ be invertible.
 - a) Prove that the eigenvalues of $AA^\top$ are positive real numbers.
 - b) Suppose there are two distinct positive integers p and q with $(AA^\top)^p = (A^\top A)^q$. Prove that $A^\top = A^{-1}$.
 2. (Romania national olympiad 2011) A row of a matrix in $M_n(\mathbb{C})$ is called permutable if the value of the determinant does not change however the entries of that row are permuted. Prove that if a matrix has two permutable rows, then its determinant equals 0.
 3. (IMC 2005) In the linear space of all real $n \times n$ matrices find the maximum possible dimension of a linear subspace V such that $\text{tr}(XY) = 0$ for all $X, Y \in V$.
 4. (Romania national olympiad 2008) Let $A \in M_n(\mathbb{R})$ satisfy $A^\top = -A$. Prove that for all non-negative reals x, y

$$\det(A + xE) \cdot \det(A + yE) \geq \det(A + \sqrt{xy} E)^2.$$

5. (Putnam 2019) Let Q be an $n \times n$ real orthogonal matrix and let $u \in \mathbb{R}^n$ be a unit column vector. Let $P = E - 2uu^\top$ be the corresponding Householder transformation. Show that if 1 is not an eigenvalue of Q , then 1 is an eigenvalue of PQ .
6. (Romania national olympiad 2019) Let $n > 2$ and let $A, B \in M_n(\mathbb{C})$ be such that there is an idempotent $C \in M_n(\mathbb{C})$ with $\text{adj } C = AB - BA$. Prove that $(AB - BA)^2 = O$.
7. (IMC) Let $\alpha \in \mathbb{R} \setminus \{0\}$ and let F, G be linear operators on $\mathbb{R}^n$ with $F \circ G - G \circ F = \alpha F$.
 - a) Show that $F^k \circ G - G \circ F^k = \alpha k F^k$ for all $k \in \mathbb{N}$.
 - b) Show that $F^k = 0$ for some $k \geq 1$.

8. (ФКН, осень 2025) Does there exist a 3×3 matrix X over $\mathbb{C}$ such that

$$X^2 - X \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} ?$$